1.6.4. The Promotion from *Testing* to *Stable*

Let us suppose that our package is now included in *Testing*. As long as it has room for improvement, its maintainer must continue to improve it and restart the process from *Unstable* (but its later inclusion in *Testing* is generally faster: unless it changed significantly, all of its dependencies are already available). When it reaches perfection, the maintainer has completed their work. The next step is the inclusion in the *Stable* distribution, which is, in reality, a simple copy of *Testing* at a moment chosen by the Release Manager. Ideally, this decision is made when the installer is ready, and when no program in *Testing* has any known critical bugs.

Since this moment never truly arrives, in practice, Debian must compromise: remove packages whose maintainer has failed to correct bugs on time, or agree to release a distribution with some bugs in the thousands of programs. The Release Manager will have previously announced a freeze period, during which each update to *Testing* must be approved. The goal here is to prevent any new version (and its new bugs), and to only approve updates fixing bugs.

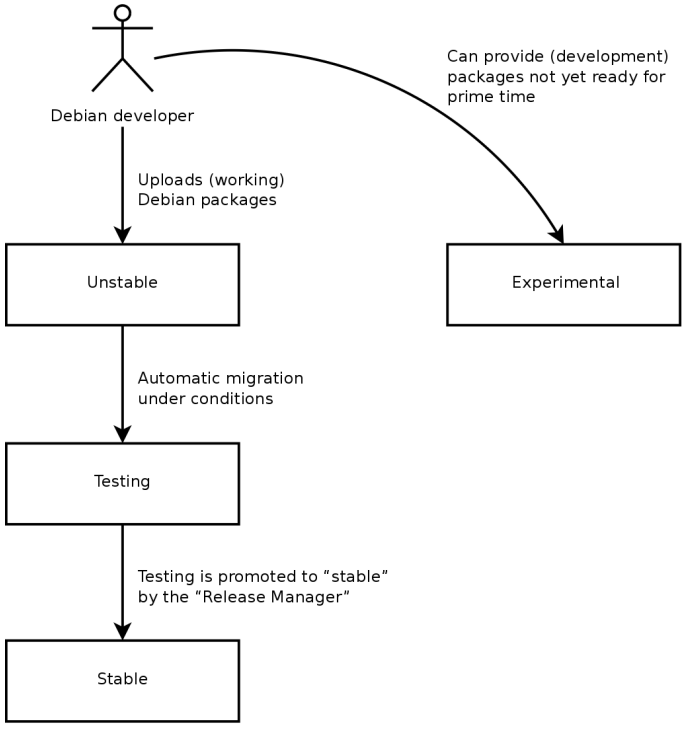


Figure 1.3 *A package's path through the various Debian versions*

VOCABULARY

Freeze: the home straight

During the freeze period, development of the *Testing* distribution is blocked; no more automatic updates are allowed. Only the Release Managers are then authorized to change packages, according to their own criteria. The purpose is to prevent the appearance of new bugs by introducing new versions; only thoroughly examined updates are authorized when they correct significant bugs.

After the release of a new stable version, the Stable Release Managers manage all further development (called "revisions", ex: 7.1, 7.2, 7.3 for version 7). These updates systematically include all security patches. They will also include the most important corrections (the maintainer of a package must prove the gravity of the problem that they wish to correct in order to have their updates included).

At the end of the journey, our hypothetical package is now included in the stable distribution. This journey, not without its difficulties, explains the significant delays separating the Debian Stable releases. This contributes, over all, to its reputation for quality. Furthermore, the majority of users are satisfied using one of the three distributions simultaneously available. The system administrators, concerned above all about the stability of their servers, don't need the latest and greatest version of GNOME; they can choose Debian *Stable*, and they will be satisfied. End users, more interested in the latest versions of GNOME or KDE Plasma than in rock-solid stability, will find Debian *Testing* to be a good compromise between a lack of serious problems and relatively up-to-date software. Finally, developers and more experienced users may blaze the trail, testing all the latest developments in Debian *Unstable* right out of the gate, at the risk of suffering the headaches and bugs inherent in any new version of a program. To each their own Debian!

CULTURE

GNOME and KDE Plasma, graphical desktop environments

GNOME (GNU Network Object Model Environment) and Plasma by KDE (formerly known as K Desktop Environment) are the two most popular graphical desktop environments in the free software world, and will be presented in greater detail in section 13.3, "Graphical Desktops" page 385.

A desktop environment is a set of programs grouped together to allow easy management of the most common operations through a graphical interface. They generally include a file manager, office suite, web browser, e-mail program, multimedia accessories, etc. The most visible difference resides in the choice of the graphical library used: GNOME has chosen GTK+ (free software licensed under the LGPL), and the KDE community has selected Qt (a company-backed project, available nowadays both under the GPL and a commercial license).

➨ `https://www.gnome.org/`

➨ `https://www.kde.org/`

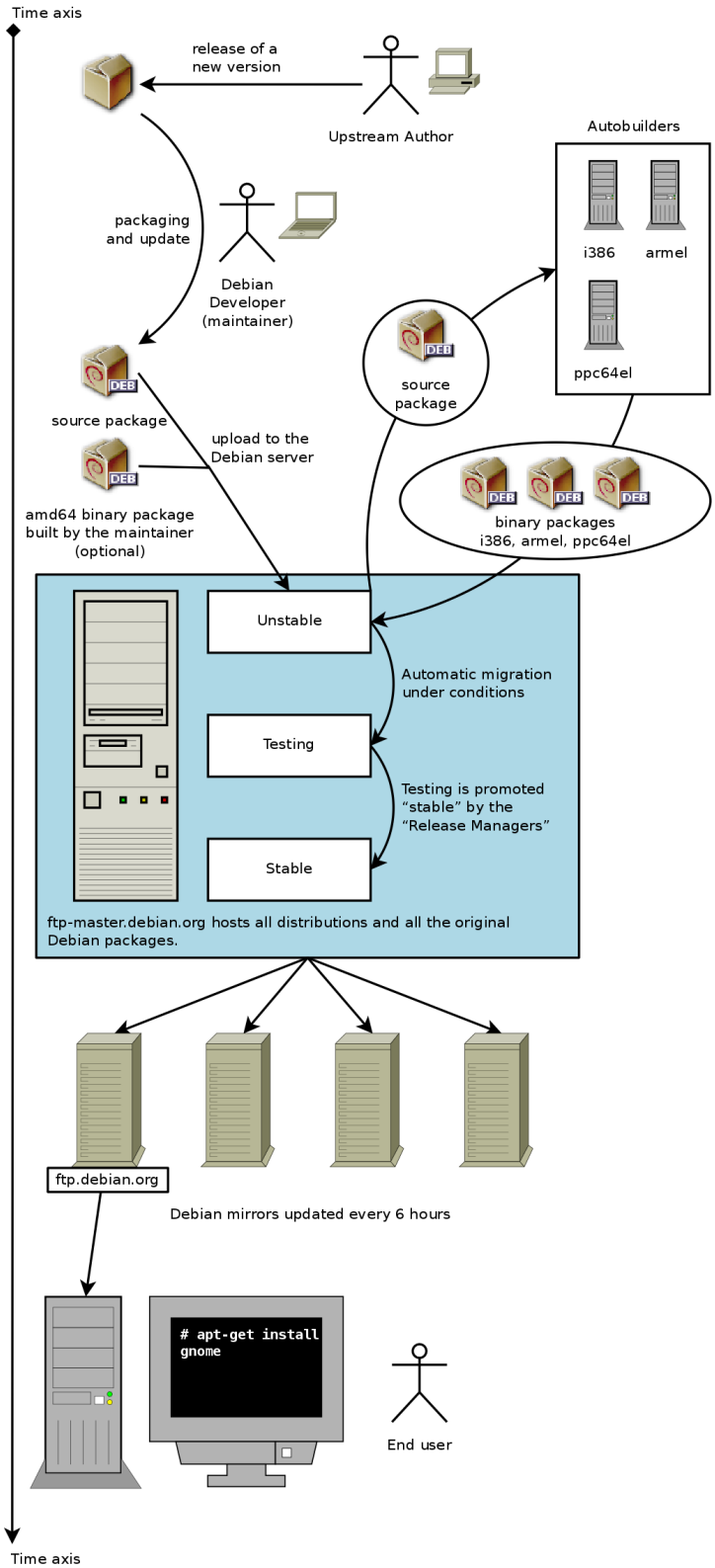


Figure 1.4 *Chronological path of a program packaged by Debian*

1.6.5. The *Oldstable* and *Oldoldstable* Status

Each *Stable* release has an expected lifetime of about 5 years and given that releases tend to happen every 2 years, there can be up to 3 supported releases at a given point of time. When a new stable release happens, the former release becomes *Oldstable* and the one even before becomes *Oldoldstable*.

This Long Term Support (LTS) of Debian releases is a recent initiative: individual contributors and companies joined forces to create the Debian LTS team. Older releases which are no longer supported by the Debian security team fall under the responsibility of this new team.

The Debian security team handles security support in the current *Stable* release and also in the *Oldstable* release (but only for as long as is needed to ensure one year of overlap with the current stable release). This amounts roughly to three years of support for each release. The Debian LTS team handles the last (two) years of security support so that each releases benefits from at least 5 years of support and so that users can upgrade from version N to N+2, for example from Debian 8 "Jessie" to Debian 10 "Buster".

➨ `https://wiki.debian.org/LTS`

COMMUNITY

Companies sponsoring the LTS effort

Long Term Support is a difficult commitment to make in Debian because volunteers tend to avoid the work that is not very fun. And providing security support for 5 years old software is — for many contributors — a lot less fun than packaging new upstream versions or developing new features.

To bring this project to life, the project counted on the fact that long term support was particularly relevant for companies and that they would be willing to mutualize the cost of this security support.

The project started in June 2014: some organizations allowed their employees to contribute part-time to Debian LTS while others preferred to sponsor the project with money so that Debian contributors get paid to do the work that they would not do for free. Most Debian contributors willing to be paid to work on LTS got together to create a clear sponsorship offer managed by Freexian (Raphaël Hertzog's company):

➨ `https://www.freexian.com/services/debian-lts.html`

In the Debian LTS team, the volunteers work on packages they care about while the paid contributors prioritize packages used by their sponsors.

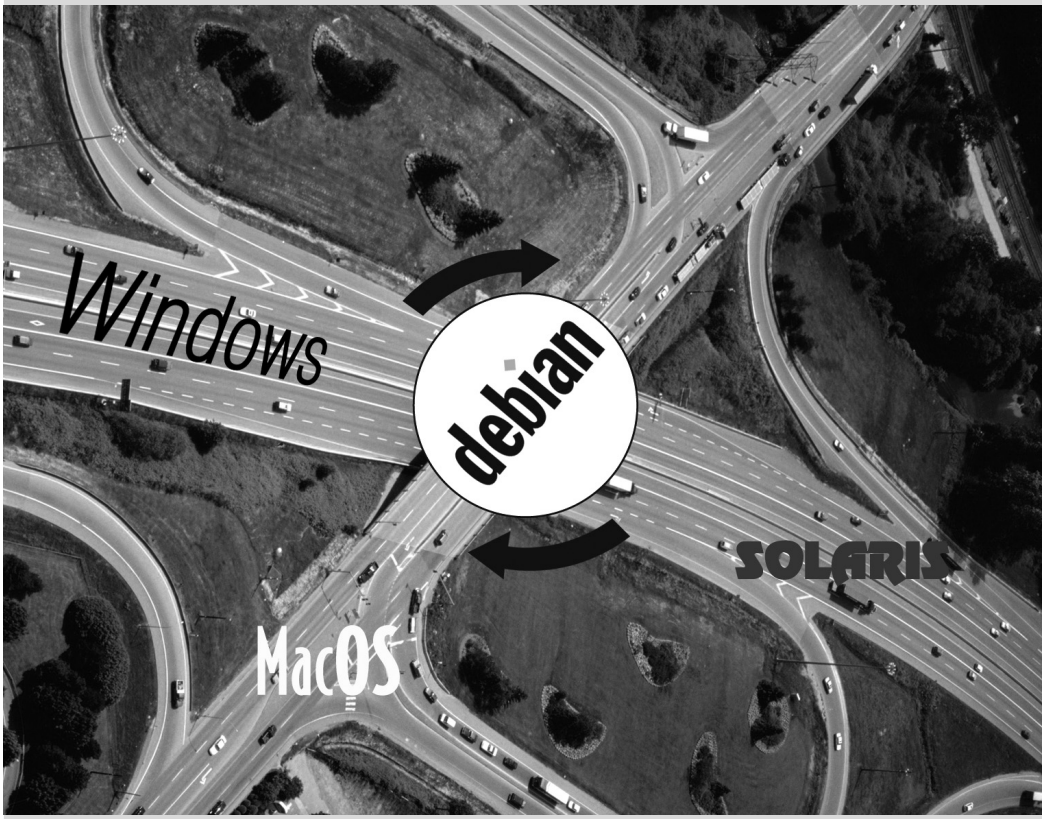
The project is always looking for new sponsors: What about your company? Can you let an employee work part-time on long term support? Can you allocate a small budget for security support?

➨ `https://wiki.debian.org/LTS/Funding`



Keywords

Falcot Corp
SMB
Strong Growth
Master Plan
Migration
Cost Reduction



Chapter

2

Presenting the Case Study

Contents

Fast Growing IT Needs 34 Master Plan 34 Why a GNU/Linux Distribution? 35
Why the Debian Distribution? 37 Why Debian Buster? 38

In the context of this book, you are the system administrator of a growing small business. The time has come for you to redefine the information systems master plan for the coming year in collaboration with your directors. You choose to progressively migrate to Debian, both for practical and economical reasons. Let's see in more detail what is in store for you...

We have envisioned this case study to approach all modern information system services currently used in a medium sized company. After reading this book, you will have all of the elements necessary to install Debian on your servers and fly on your own wings. You will also learn how to efficiently find information in the event of difficulties.

2.1. Fast Growing IT Needs

Falcot Corp is a manufacturer of high quality audio equipment. The company is growing strongly, and has two facilities, one in Saint-Étienne, and another in Montpellier. The former has around 150 employees; it hosts a factory for the manufacturing of speakers, a design lab, and all administrative office. The Montpellier site is smaller, with only about 50 workers, and produces amplifiers.

NOTE **Fictional company created for case study**

The Falcot Corp company used as an example here is completely fictional. Any resemblance to an existing company is purely coincidental. Likewise, some example data throughout this book may be fictional.

The information system has had difficulty keeping up with the company's growth, so they are now determined to completely redefine it to meet various goals established by management:

- modern, easily scalable infrastructure;
- reducing cost of software licenses thanks to use of Open Source software;
- installation of an e-commerce website, possibly B2B (business to business, i.e. linking of information systems between different companies, such as a supplier and its clients);
- significant improvement in security to better protect trade secrets related to new products.

The entire information system will be overhauled with these goals in mind.

2.2. Master Plan

With your collaboration, IT management has conducted a slightly more extensive study, identifying some constraints and defining a plan for migration to the chosen Open Source system, Debian.

A significant constraint identified is that the accounting department uses specific software, which only runs on Microsoft Windows™. The laboratory, for its part, uses computer aided design software that runs on OS X™.

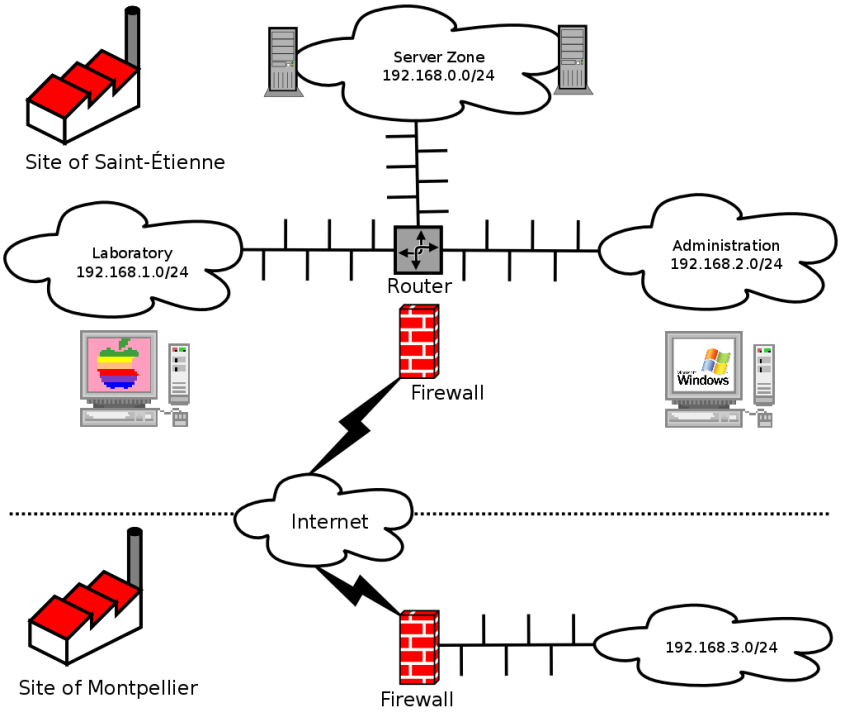


Figure 2.1 *Overview of the Falcot Corp network*

The switch to Debian will be gradual; a small business, with limited means, cannot reasonably change everything overnight. For starters, the IT staff must be trained in Debian administration. The servers will then be converted, starting with the network infrastructure (routers, firewalls, etc.) followed by the user services (file sharing, Web, SMTP, etc.). Then the office computers will be gradually migrated to Debian, for each department to be trained (internally) during the deployment of the new system.

2.3. Why a GNU/Linux Distribution?

BACK TO BASICS

Linux or GNU/Linux?

Linux, as you already know, is only a kernel. The expressions, "Linux distribution" and "Linux system" are, thus, incorrect: they are, in reality, distributions or systems *based on* Linux. These expressions fail to mention the software that always completes this kernel, among which are the programs developed by the GNU Project. Dr. Richard Stallman, founder of this project, insists that the expression "GNU/Linux" be systematically used, in order to better recognize the important contributions made by the GNU Project and the principles of freedom upon which they are founded.

Debian has chosen to follow this recommendation, and, thus, name its distributions accordingly (thus, the latest stable release is Debian GNU/Linux 10).

Several factors have dictated this choice. The system administrator, who was familiar with this distribution, ensured it was listed among the candidates for the computer system overhaul. Difficult economic conditions and ferocious competition have limited the budget for this operation, despite its critical importance for the future of the company. This is why Open Source solutions were swiftly chosen: several recent studies indicate they are less expensive than proprietary solutions while providing equal or better quality of service so long as qualified personnel are available to run them.

IN PRACTICE

Total cost of ownership (TCO)

The Total Cost of Ownership is the total of all money expended for the possession or acquisition of an item, in this case referring to the operating system. This price includes any possible license fee, costs for training personnel to work with the new software, replacement of machines that are too slow, additional repairs, etc. Everything arising directly from the initial choice is taken into account.

This TCO, which varies according to the criteria chosen in the assessment thereof, is rarely significant when taken in isolation. However, it is very interesting to compare TCOs for different options if they are calculated according to the same rules. This assessment table is, thus, of paramount importance, and it is easy to manipulate it in order to draw a predefined conclusion. Thus, the TCO for a single machine doesn't make sense, since the cost of an administrator is also reflected in the total number of machines they manage, a number which obviously depends on the operating system and tools proposed.

Among free operating systems, the IT department looked at the free BSD systems (OpenBSD, FreeBSD, and NetBSD), GNU Hurd, and Linux distributions. GNU Hurd, which has not yet released a stable version, was immediately rejected. The choice is simpler between BSD and Linux. The former have many merits, especially on servers. Pragmatism, however, led to choosing a Linux system, since its installed base and popularity are both very significant and have many positive consequences. One of these consequences is that it is easier to find qualified personnel to administer Linux machines than technicians experienced with BSD. Furthermore, Linux adapts to newer hardware faster than BSD (although they are often neck and neck in this race). Finally, Linux distributions are often more adapted to user-friendly graphical user interfaces, indispensable for beginners during migration of all office machines to a new system.

ALTERNATIVE

Debian GNU/kFreeBSD

Since Debian 6 *Squeeze*, it is possible to use Debian with a FreeBSD kernel on 32 and 64 bit computers; this is what the `kfreebsd-i386` and `kfreebsd-amd64` architectures mean. While these architectures are not "official" (they are hosted on a separate mirror — `ports.debian.org`), they provide over 70% of the software packaged by Debian.

These architectures may be an appropriate choice for Falcot Corp administrators, especially for a firewall (the kernel supports three different firewalls: IPF, IPFW, PF) or for a NAS (network attached storage system, for which the ZFS filesystem has been tested and approved).

2.4. Why the Debian Distribution?

Once the Linux family has been selected, a more specific option must be chosen. Again, there are plenty of criteria to consider. The chosen distribution must be able to operate for several years, since the migration from one to another would entail additional costs (although less than if the migration were between two totally different operating systems, such as Windows or OS X).

Sustainability is, thus, essential, and it must guarantee regular updates and security patches over several years. The timing of updates is also significant, since, with so many machines to manage, Falcot Corp can not handle this complex operation too frequently. The IT department, therefore, insists on running the latest stable version of the distribution, benefiting from the best technical assistance, and guaranteed security patches. In effect, security updates are generally only guaranteed for a limited duration on older versions of a distribution.

Finally, for reasons of homogeneity and ease of administration, the same distribution must run on all the servers and office computers.

2.4.1. Commercial and Community Driven Distributions

There are two main categories of Linux distributions: commercial and community driven. The former, developed by companies, are sold with commercial support services. The latter are developed according to the same open development model as the free software of which they are comprised.

A commercial distribution will have, thus, a tendency to release new versions more frequently, in order to better market updates and associated services. Their future is directly connected to the commercial success of their company, and many have already disappeared (Caldera Linux, StormLinux, Mandriva Linux, etc.).

A community distribution doesn't follow any schedule but its own. Like the Linux kernel, new versions are released when they are stable, never before. Its survival is guaranteed, as long as it has enough individual developers or third party companies to support it.

A comparison of various Linux distributions led to the choice of Debian for various reasons:

- It is a community distribution, with development ensured independently from any commercial constraints; its objectives are, thus, essentially of a technical nature, which seem to favor the overall quality of the product.
- Of all community distributions, it is the most significant from many perspectives: in number of contributors, number of software packages available, and years of continuous existence. The size of its community is an incontestable witness to its continuity.
- Statistically, new versions are released every 18 to 24 months, and they are supported for 5 years, a schedule which is agreeable to administrators.
- A survey of several French service companies specialized in free software has shown that all of them provide technical assistance for Debian; it is also, for many of them, their

chosen distribution, internally. This diversity of potential providers is a major asset for Falcot Corp's independence.

- Finally, Debian is available on a multitude of architectures, including ppc64el for OpenPOWER processors; it will, thus, be possible to install it on Falcot Corp's latest IBM servers.

IN PRACTICE

Debian Long Term Support

The Debian Long Term Support (LTS) project started in 2014 and aims to provide 5 years of security support to all stable Debian releases. As LTS is of primary importance to organizations with large deployments, the project tries to pool resources from Debian-using companies.

➨ https://wiki.debian.org/LTS

Falcot Corp is not big enough to let one member of its IT staff contribute to the LTS project, so the company opted to subscribe to Freexian's Debian LTS contract and provides financial support. Thanks to this, the Falcot administrators know that the packages they use will be handled in priority and they have a direct contact with the LTS team in case of problems.

➨ https://wiki.debian.org/LTS/Funding

➨ https://www.freexian.com/services/debian-lts.html

Once Debian has been chosen, the matter of which version to use must be decided. Let us see why the administrators have picked Debian *Buster*.

2.5. Why Debian Buster?

Every Debian release starts its life as a continuously changing distribution, also known as "*Testing*". But at the time you read those lines, Debian *Buster* is the latest "*Stable*" version of Debian.

The choice of Debian *Buster* is well justified based on the fact that any administrator concerned about the quality of their servers will naturally gravitate towards the stable version of Debian. Even if the previous stable release might still be supported for a while, Falcot administrators aren't considering it because its support period will not last long enough and because the latest version brings new interesting features that they care about.

